Homework 4

INFO 5265: Accessibility and Inclusive Design

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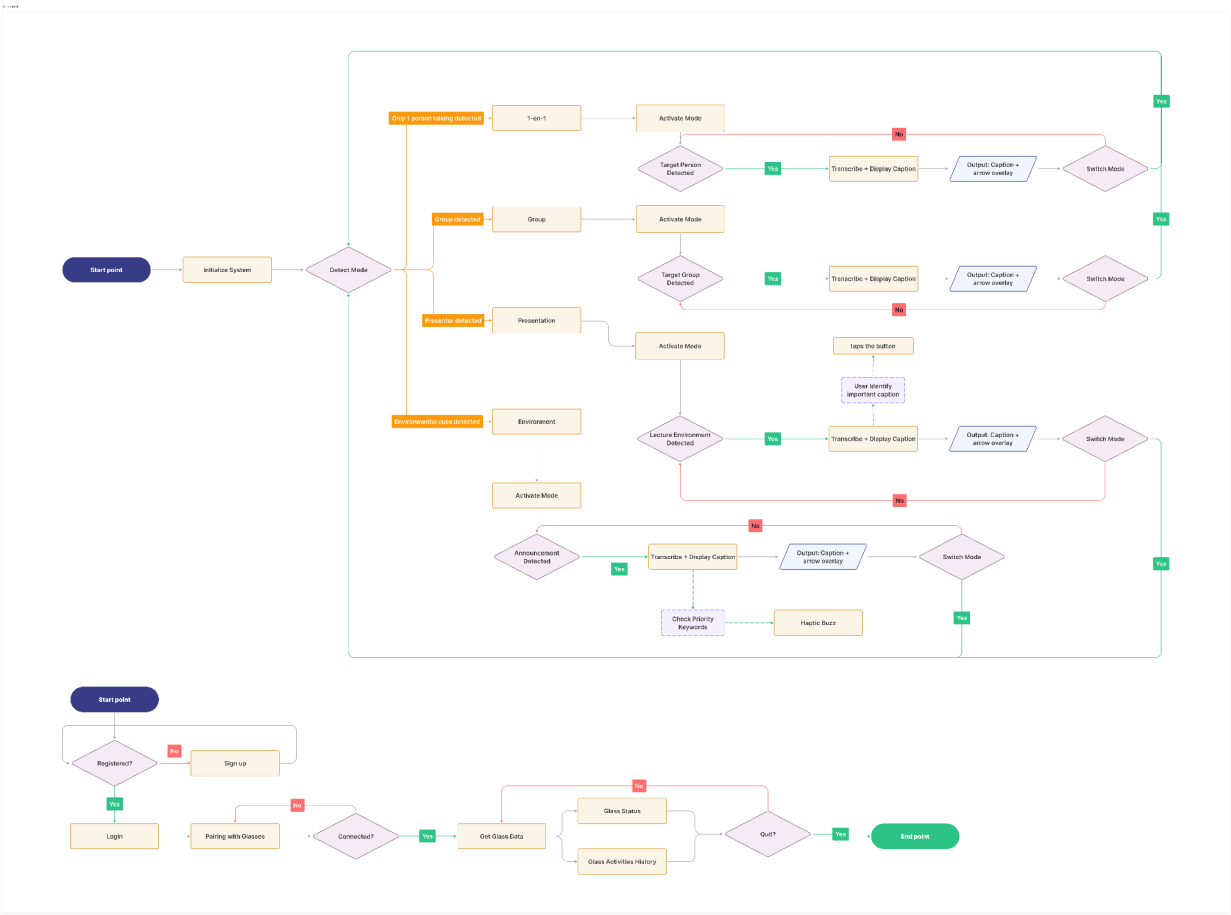
Part 1: Horizontal Prototype

Our horizontal prototype illustrates the breadth of our smart glasses system designed for people with hearing loss. It represents the overall user experience across both the companion mobile application and the smart glasses interface. The prototype focuses on how users move between major states of the system.

The flow begins when a user opens the companion app, signs in, and pairs their glasses. After successful pairing, they reach the Home Screen, which provides access to four key modes: 1-on-1 Conversation, Group Discussion, Presentation/Class, and Environmental Detection. Each mode automatically detects the surrounding sound context and adjusts caption placement, color-coding, and visual emphasis accordingly. Users can also navigate back to the home screen or settings at any time.



This flowchart shows how the system balances automation (e.g., auto-detecting sound types and adjusting display style) with optional user interaction (e.g., selecting modes, pinning a speaker, reviewing transcripts). The companion app provides setup and navigation, while the glasses deliver real-time visual feedback.

Part 2: Vertical Prototype

Prototype Demo

https://docs.google.com/presentation/d/1zQMky3gbb70EKxbb9QzADEiU-xyw8oQlHgLWIRKMrdE/edit?slide=id.g39d9c909482_2_32#slide=id.g39d9c909482_2_32

Story Context

Jiawen and Iris are working on a group assignment in the **Bloomberg Level 0 study area**. It's noisy, with many students passing by. In this scenario, Jiawen is the Glass wearer.
The glasses automatically recognize the **target person (Iris)** and display captions for Iris' speech for Jiawen. *(Feature 1: Real-time Captioning)*

Later, Yuhan and August join in. They greet Jiawen and Iris and sit down to discuss homework assignment, shifting into **Group Conversation Mode**. *(Feature 1: Real-time Captioning)*

Later, they attend the **lecture**, where the professor Shiri speaks, Jiawen takes notes, and another person (off-screen, in this scenario is Ricky) asks a question. The system detects the **sound direction** and dynamically adjusts the caption placement. *(Feature 2: Directional Audio Mapping)*

Finally, the **fire alarm** goes off, triggering the **Emergency Alert Overlay (Environmental Mode)**. Everyone leaves safely. *(Feature 3: Emergency Visualization)*

Feature 1: Live Captioning

Key Question:
How can captions remain clear, natural, and easy to follow during both one-on-one and group conversations in noisy or dynamic environments?

Alternative Designs

1. **Design A: Central Caption Bar**
    - **Description:** A floating caption bar appears near the bottom center of the lens, showing transcribed speech in distinct colors for each speaker. Text fades out automatically after a few seconds.
    - **User Input:** None, captions are generated through automatic speech detection.
    - **System Output**: Real-time text appears in a clear, color-coded font on a translucent background (e.g., blue for Iris, yellow for August).
    - **Rationale:** The central placement keeps text within the user's natural line of sight, providing fast readability and minimal eye movement. Ideal for one-on-one or small group settings.
2. **Design B: Speak Bubble**
    - **Description**: Each speaker's speech appears as a colored text bubble positioned near their face. When multiple people speak, each bubble remains anchored to the speaker's location and identity.
    - **User Input:** Optional tap to confirm or pin a primary speaker.

- **System Output:** A colored bubble (e.g., purple caption bubble for Yuhan, orange caption bubble for Ricky) displays text above or next to the speaker's head, synchronized with their voice.
- **Rationale:** By linking speech directly to the visual position of each speaker, this design strengthens spatial association and social connection, helping users immediately identify who is talking in group settings.

3. **Design C: Side "Karaoke-Style" Chatbox**
    - **Description:** A vertical chat panel appears along one edge of the lens, continuously displaying dialogue in order of speech. The current speaker's line is highlighted, while previous captions fade slightly (Samaradivakara et al., 2025).
    - **User Input:** Swipe gesture to toggle visibility of the chatbox.
    - **System Output:** A scrolling list of color-coded captions (e.g., latest line in bright color, previous lines dimmed).
    - **Rationale:**This design emphasizes temporal flow and context retention, allowing users to glance back at recent dialogue while maintaining awareness of the current conversation. It works best for longer, structured discussions such as meetings or group work.

Feature 2: Directional Mapping

Key Question:
How can captions be spatially aligned with sound direction when the speaker is off-screen?

Alternative Designs

1. **Design A: Arrow Pointer + Caption**
    - **Description:** When a speaker talks outside the user's field of view, an arrow appears on the edge of the display, pointing toward the direction of the sound. The caption text appears near the arrow, showing what the speaker says.
    - **System Output:** When off-screen speech is detected, the system displays an arrow and short caption ("The next important theory is →") aligned toward the speaker's direction.
    - **Rationale:** Provides clear directional feedback that helps the user locate who is speaking. The arrow acts as a visual cue for orientation while keeping the caption readable and anchored to the sound source.
2. **Design B: Directional Icon Indicator**
    - **Description:** A sound-direction icon (e.g., a pen or wave symbol) appears near the caption text to indicate the speaker's relative position. The caption remains centered in the user's view while the icon subtly signals direction.
    - **System Output:** When the system detects speech from the side or behind, it shows a caption ("The next important theory is") paired with a directional icon on one side, indicating where the voice originates.
    - **Rationale:** Offers a lightweight visual cue without shifting the caption position. Ideal for lectures or note-taking situations where maintaining text focus is more important than precise localization.
3. **Design C: 360° Compass Interface**
    - **Description:** A semi-transparent compass overlay appears at the center or top of the display, showing N–E–S–W directions. Sound sources are marked dynamically on the compass, and captions appear adjacent to the active direction.
    - **System Output:** When multiple speakers are detected (e.g., during a class discussion), the compass highlights active directions (e.g., "E" for a voice from

the right) and places captions such as “The next important concept is” next to that marker.
    - **Rationale:** Supports spatial awareness in environments with multiple sound sources. Helps the user distinguish between voices from different directions without turning their head, suitable for group or classroom contexts.

Feature 3: Environment Detection

Key Question:
What is the most effective way to visualize environmental emergencies without overwhelming the user?

Alternative Designs

1. **Design A: Full Red Tint Alert**
    - When an emergency such as a fire alarm is detected, the entire lens view gradually transitions into a **semi-transparent red tint**. A bold alert message (e.g., *“A fire emergency has been reported”*) appears at the bottom of the screen.
    - **User Input:** None, the system automatically activates upon detecting a flashing alarm or emergency signal.
    - **System Output:** A bright red overlay fills the lens, accompanied by high-contrast white or black text in large type.All other captions and UI elements are temporarily hidden to focus attention on the alert.
    - **Rationale:**The strong contrast and immersive tint ensure **instant recognition** of critical danger.This design prioritizes **speed and clarity** in urgent scenarios where every second counts.
2. **Design B: Central Alert Label**
    - **Description:** A **large alert label** appears prominently in the middle of the display without changing the full lens color. The text (e.g., *“Fire Alert”*) is framed in a semi-transparent box with a glowing border to distinguish it from background visuals.
    - **User Input:** None, activation is automatic once environmental sensors detect an alarm.
    - **System Output:** An animated label pulses gently in brightness while keeping the rest of the view visible, allowing the user to stay oriented within the environment.
    - **Rationale:** This approach balances **visibility and situational awareness**, ensuring the user notices the alert while still perceiving obstacles and exit routes clearly.
3. **Design C: Red Dashed Line Indicator**
    - **Description:** A subtle red dashed outline appears along the edges of the lens, pulsing softly to indicate emergency mode. A brief vibration cue accompanies the visual signal to reach users with hearing impairments.
    - **User Input:** None, system triggers automatically via environmental sensors.
    - **System Output:** The pulsing border glows in rhythm with the detected alarm sound. The central view remains fully visible for navigation, and the user feels a short haptic pulse through the frame.
    - **Rationale:** This design is optimized for accessibility and calm response.
      It preserves full visibility of the environment while providing clear, multimodal feedback, visual, spatial, and tactile, for immediate awareness.

Part 3: Reflect

Through building our horizontal and vertical prototypes, we gained a concrete understanding of how smart glasses could adapt caption displays to different social and environmental contexts for people with hearing loss. Initially, we assumed full automation. The system detects, transcribes, and captions everything without user input which would create the most seamless experience. However, testing revealed that this approach sometimes reduced users' sense of control. Allowing lightweight interaction, such as pinning a primary speaker or toggling caption visibility, gave users more confidence and flexibility. By comparing three caption layouts (the central bar, anchored speech bubbles, and side chatbox). We learned that readability depends not only on text clarity but also on visual stability and eye movement within the AR space. The central bar worked best for simple one-on-one conversations, while speech bubbles provided useful spatial cues in small groups but risked visual clutter. Directional mapping features such as arrows and compass overlays proved highly effective for maintaining spatial awareness when speakers were off-screen, helping users orient themselves in noisy or dynamic environments. Prototyping the environmental alert system further highlighted the importance of balancing visibility and safety: a full red overlay immediately drew attention but obstructed awareness, whereas a subtle red outline with vibration offered an accessible yet non-intrusive signal.

Looking forward, we plan to extend our design by incorporating ideas from Augmenting Captions with Emotional Cues: An AR Interface for Real-Time Accessible Communication (Ubur, 2025). Our next step is to explore how emotional and non-verbal cues (such as tone or emphasis can be visually represented through color, iconography, or subtle motion, making captions not only informative but also empathetic.) We also aim to develop adaptive display behaviors that respond to speech intensity and user attention, as well as a user feedback loop to personalize readability and emotional accuracy. This direction will move our project beyond pure transcription toward designing a communication system that helps users understand not just the words being spoken, but the emotions and intent behind them.

References

Samaradivakara, Y., Pathirage, A., Ushan, T., Sasikumar, P., Karunanayaka, K., Keppitiyagama, C., & Nanayakkara, S. (2025). Tailored Real-time AR Captioning Interface for Enhancing Learning Experience of Deaf and Hard-of-Hearing (DHH) Students. *ArXiv*, abs/2501.02233.

Ubur, S.D. (2025). Augmenting Captions with Emotional Cues: An AR Interface for Real-Time Accessible Communication. *ArXiv*, abs/2504.17171.